Structure and Data: QCA as a Framework for Separating Emergent Structure and Contingent Parameters in Particle Physics

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Abstract

Central thesis: Particle physics splits into two distinct classes — *structure*, which can be derived from the local unitary rules of a quantum cellular automaton (QCA), and *data* (parameters), which must be assigned externally. We test this thesis numerically. The *structure* — encompassing the Dirac equation, gauge symmetries, the Higgs mechanism, and the mechanism generating three fermion generations — is reproduced from QCA rules with an accuracy of 10^{-15}–10^{-17} (8 distinct structural sectors, Class A). Consistency checks with known quantum mechanical theorems (CPT, $c = 1$, entanglement, Pauli exclusion; Class B) and negative controls (square lattice, fundamental constants, CP conservation in 1D and 2 generations; Class C) define the boundary of this classification. Conversely, *parameters* (masses, Yukawa couplings, fundamental constants, CP phase) cannot be derived and constitute input data (4 data classifications). This work makes **no new physical predictions**; its contribution is strictly methodological: it provides a rigorous framework for classifying what is derivable from local unitarity and what must remain as input data. We additionally investigate a base-15 hypothesis for lepton masses. Using pairwise logarithmic ratios m_j/m_i, base 15 yields a deviation of 0.028148 for leptons. A permutation test over 84 partitions of the 9 masses gives $p_{abs} = 10/84 = 0.1190$; the status is **not established** (does not reach the 5% significance level). The look-elsewhere effect is prominent: the triplet $\{\mu, b, t\}$ at base 40 yields a deviation of 0.005912, which is 4.76 times better than the lepton signal. For quarks, no base yields a deviation below 5%. The measured value $\Sigma m_\nu = 0.0608$ eV (nb13) is a consistency check with the DESI DR2 upper bound. The framework does not depend on the base-15 hypothesis.

1 Introduction

1.1 The Epistemological Paradox

The project began with a question about artificial intelligence. There exists a two-way epistemological wall between us and AI. The inversion of this observation is: *what if we ourselves are in the position of an AI relative to some substrate?* We see the "output" (observable physics), but we do not see the "hardware" (the substrate).

1.2 Analogy: "Arithmetic and Ink"

Imagine a sheet of paper with numbers written on it. You are a mathematician who knows only arithmetic and geometry. You can add numbers and measure the distances between them. But you **cannot** deduce from arithmetic why the digit "7" is drawn with straight lines while "2" is curved, or why "5" is written in red and "7" in blue. These properties are **real**, but they are

not part of the mathematical rules. They are **characteristics of the medium** (the ink, the handwriting).

In this project, we show that physics is structured analogously: **rules** (symmetries, interactions, the speed of light, CPT, entanglement, three generations) are derivable, whereas **parameters** (masses, coupling constants, the CP phase) are the "ink" that cannot be derived from the rules.

It is worth noting how this approach differs from existing programs. The Froggatt–Nielsen mechanism [13] attempts to explain the mass hierarchy via new symmetries (which would make masses *structure*), while Wolfram's computational model [14] treats the universe as a network of rules (similar to our QCA ontology). Our approach differs: we do not attempt to derive the parameters (Froggatt–Nielsen), nor do we claim the hypergraph generates specific particle masses (Wolfram). Instead, we classify which parts of the Standard Model *can* be derived and rigorously isolate the rest as irreducible input data.

1.3 What This Approach Does *Not* Claim

We emphasize three points to prevent misunderstanding.

First, the approach does **not** claim that the Universe is a *simulation*. There is no "simulator outside" executing our world as a program. QCA is an **ontology**, not the simulation hypothesis: the claim is that physics *is* a local unitary rule, not that someone *computes* it. There is no "before the rule" and no "outside the rule" — there is only the rule. This distinction aligns with the information-theoretic view of the universe [9], but it strictly avoids the implication of an external simulator. It is also closely related to the ontological interpretation of cellular automata by 't Hooft [10] and Elze [11], who likewise treat the automaton as the fundamental rule rather than as a computational simulation of external hardware.

Second, the approach does **not** claim to be a "theory of everything". It classifies what is derivable from local unitarity and what must be given as input; it does not derive the Standard Model, gravity, or cosmology. Eight distinct sectors of Class A, four checks of Class B, five negative controls of Class C, and four data classifications — this is the **full scope** of the work, nothing more.

Third, the approach does **not** claim that the substrate is directly knowable. We see the "output" (physics) and classify it into structure and data. We do not see the hardware level. Whether the substrate exists, what it is, and how our output relates to it — the approach **leaves in brackets** and does not resolve. The structure/data classification works *inside* the output, regardless of what lies behind it.

2 Methodology

Quantum Cellular Automaton (QCA) is a minimal model of local unitary computation: a discrete lattice, discrete time steps, and a local update rule $\mathcal{U}$. It represents the smallest set of assumptions required to ask: "What follows from locality and unitarity?" For a comprehensive overview of the QCA framework, including its structural and universality results, see Arrighi [12].

Relationship to previous work. The derivations of Weyl, Dirac, and Maxwell equations from QCA are not new: they were established analytically by the Pisa group (D'Ariano, Perinotti, Bisio, and Tosini) [8, 7]. Our contribution is not the re-derivation of these known results, but the construction of a unified **classification framework** (8+4+5+4). Specifically, we provide: (i) the architectural separation of structure and data; (ii) the negative controls (Class C) that define the boundaries of the QCA derivations; and (iii) the rigorous statistical treatment of the data. We employ the QCA as a numerical testbed to verify these established analytical results (nb01–nb03, nb06) and extend them to new sectors (nb04, nb05, nb08, nb18).

Reconstruction vs. Derivation. We acknowledge that the QCA update rules are constructed

with the target continuum limit in mind; in this sense, the procedure is a *reconstruction* rather than a blind derivation. However, this reconstruction is justified by **uniqueness theorems** established in the QCA framework [12, 8]: given locality, unitarity, and isotropy, the only possible free-field continuum limits are the Weyl, Dirac, and Maxwell equations. Therefore, the numerical reconstructions in nb01–nb03 verify these uniqueness classes. A sector is admitted to Class A only if the continuum limit is reached without fitting any parameters. The varying precision across sectors (10^{-17} for gauge fields vs. 0.1% for the Higgs BCS condensate) reflects the different numerical methods (exact diagonalization vs. iterative mean-field).

3 Structure Sectors (Class A)

There are **8 distinct sectors** of structure derived from local QCA rules:

- **nb01:** 1D Dirac equation (10^{-15});
- **nb02:** U(1) gauge field / photon (10^{-17});
- **nb03:** SU(2) / W, Z bosons (10^{-16});
- **nb04:** Independence of gauge invariance and unitarity from mass ($m \in [0.01, 1.5]$; scatter of residuals and variance below the required order);
- **nb05:** Higgs mechanism via a BCS condensate (0.1%) (within a BCS-condensate toy model; not the full SM Higgs);
- **nb06:** 2D Dirac equation on a honeycomb lattice (Fermi velocity ratio $\alpha = 1.01$, isotropy 1.0000; $\alpha \to 1$ in the continuum limit);
- **nb08:** 3D Dirac on a 4D domain wall ($E_0 = 0$, probability density on the wall 51%);
- **nb18 Parts B–C:** Three generations — N walls $\to$ N modes in 1D SSH (exact correspondence, $|\langle\gamma_5\rangle| = 1.0000$); 2D is realized via overlap fermions (Ginsparg–Wilson residual $\sim 3 \times 10^{-15}$).

3.1 Consistency Checks (Class B)

These are not "confirmation levels" in the same sense — QCA reproduces known quantum mechanical theorems without adding independent content:

- **nb12:** CPT, $m = \bar{m}$;
- **nb15:** $c = 1$ from locality (group velocity $v_F = 1.000000$ in step/tick units);
- **nb16:** CHSH $= 2\sqrt{2}$ is preserved;
- **nb17:** Exchange sign -1 for fermions, $+1$ for bosons.

3.2 Negative Controls (Class C)

- **nb07:** Square 2D lattice yields anisotropy > 1.15 (contrast with the honeycomb lattice);
- **nb11:** Null result for the 4D wall geometry (no hidden 1/32 factors in velocities);
- **nb14:** No hidden grid for fundamental constants (error $\sim 10^{-3}$);
- **nb18 Part A:** Naive 2D lattice — 4 Dirac points (fermion doubling);
- **nb19 T0–T2:** CP is conserved in 1D and 2 generations.

How negative controls were tested. All controls use the same 5% decision threshold as the leptons, although their metrics are control-specific (anisotropy for nb07, error magnitude for nb14, Dirac point count for nb18A). The 84-triplet permutation test is applied exclusively to the lepton triplet — the controls are evaluated separately and are not included in the 84 partitions. Specifically, these controls exclude: (a) **quarks** — demonstrating that the base-15 preference is not a universal property of any mass triplet (0 bases with dev $< 5\%$); (b) **nb14 (constants)** — showing that the preference is not a consequence of metric choice (no hidden grid at 10^{-3}); (c) **nb07 (square lattice)** — proving that the 2D effect is not an artifact of the honeycomb structure (anisotropy > 1.15); (d) **nb18 Part A (naive lattice)** — showing that chiral anomalies do not create an illusion of topology (4 Dirac points explicitly).

3.3 Data Classifications

- **nb09:** SVD of quark masses (rank-1, 99.7%, factor 2 — effectively reflecting the dominance of the top quark mass);
- **nb10:** Base scan (see Section 7);
- **nb13:** Neutrinos — the measured mass sum $\Sigma m_\nu = 0.0608$ eV (Planck 2018 + BAO compilation) is consistent with the DESI DR2 upper bound ($\Sigma m_\nu < 0.072$ eV at 95% CL) [6]. This is a **consistency check, not a prediction**: the model does not predict the value in advance and does not fit the parameter. There is no full base-15 grid for neutrinos.
- **nb19 T4–T5:** CP violation — the phase δ cannot be derived from the rules and is an input (see Section 6).

4 Synthesis: 8 + 4 + 5 + 4

Class	**Count**	**Notebooks**	**Role**
A — Distinct structural sectors	**8**	nb01–06, 08, 18 Parts B–C	Structure is derived
B — Consistency with quantum mechanics	**4**	nb12, 15–17	Consistent with QM theorems
C — Negative controls	**5**	nb07, 11, 14, 18 Part A, 19 T0–T2	Classification is non-trivial
Data — Parameter classifications	**4**	nb09, 10, 13, 19 T4–T5	Remains as input

Table 1: The complete scope of the framework.

The counts do not overlap; negative controls are not counted as confirmation levels. nb18 is split: Part A — Class C, Parts B–C — Class A. nb19 is split: T0–T2 — Class C, T4–T5 — Data.

5 Three Generations: A Closed Sector (nb18)

5.1 Mechanism and Number

The Jackiw–Rossi mechanism [3] is **exact in 1D**: N SSH domain walls [4] yield exactly N zero modes with $E = 0$ and $|\langle\gamma_5\rangle| = 1.0000$. It is **realizable in 2D** via overlap fermions [2]

(Ginsparg–Wilson residual $\sim 3 \times 10^{-15}$). A naive 2D lattice suffers from fermion doubling (nb18 Part A) — this is a lattice artifact, not physics.

The specific number $N = 3$ is an **input fact**: the mechanism allows any N, and the choice $N = 3$ in nature is a parameter.

5.2 Verdict

Three generations: mechanism = structure, number $N = 3$ = input. The mechanism is derived exactly in 1D and is realizable in 2D. This does not depend on the base-15 hypothesis.

6 CP Violation: A Closed Sector (nb19)

T0–T2: CP is conserved in 1D with real and complex masses, as well as in 2 generations (the phase is eliminated by a chiral rotation). T4: 3 generations with a toy CKM matrix — $J(\delta) \neq 0$, maximum at $\delta = \pi/2$, $J = 0$ at $\delta = 0$. T5: Majorana phases yield a non-zero J_M, but they manifest in lepton-number-violating processes ($0\nu\beta\beta$), not in oscillations.

CP violation = data. The framework can accommodate CP violation (via 3 generations or Majorana masses), but the phase δ is not derived from the rules.

7 Base 15: An Exploratory Observation (nb10)

7.1 Statistic

Methodological note. The base-15 analysis is presented here not as a candidate for a physical discovery, but as a **methodological demonstration** of how to rigorously test and falsify numerical hypotheses (numerology) in particle physics. The primary result of this section is the demonstration of the multiplicity correction (look-elsewhere effect) and the honest classification of a null result.

In nb10, we use the statistic of pairwise logarithmic ratios:

$$\mathrm{dev}(b) = \mathrm{mean}_{i<j} \left| \log_b(m_j/m_i) - \mathrm{round}(\log_b(m_j/m_i)) \right| . \tag{1}$$

The statistic is scale-invariant by construction. The range of bases is $b \in [2, 40]$.

7.2 Result for Leptons

Base 15 is the best for the lepton triplet: $\mathrm{dev} = 0.028148 \pm 1.61 \times 10^{-5}$ (Monte Carlo over PDG errors [5], seed 42, $N = 500$ — a stability test). It is the only base with a deviation below 5%.

7.3 Permutation Test (Best-of-39)

We consider all 84 partitions of the 9 masses into groups of $3+6$. For each group T, we compute $\mathrm{dev}_{\mathrm{best39}}(T) = \min_{b\in[2,40]} \mathrm{dev}(T, b)$. Then:

$$p_{abs} = \frac{\#\{T : \mathrm{dev}_{\mathrm{best39}}(T) \leq \mathrm{dev}_{\mathrm{best39}}(\mathrm{leptons})\}}{84}. \tag{2}$$

Result: $p_{abs} = 10/84 = 0.1190$. Status: **not established** (which does not reach the 5% significance level). We note that p_{abs} is calculated on the central values of the masses; the Monte Carlo over PDG errors affects the dev values but does not change the ranking of triplets.

We acknowledge two statistical caveats: (1) the 9 masses are not strictly exchangeable (the top quark is significantly heavier than the others), which makes the permutation null conservative; (2) we use pole masses for the top quark and $\overline{\mathrm{MS}}$ masses for the lighter quarks (standard experimental

practice, as the pole mass is measured directly for the top quark and $\overline{\text{MS}}$ masses are standard for the lighter ones), which introduces a scheme dependence. Neither caveat changes the conclusion: the lepton triplet fails to stand out significantly.

Note for the reader: The lepton triplet is not extreme in either statistic. In the p_{abs} ranking (by $\text{dev}_{\text{best39}}$), it ranks 10th out of 84: 9 triplets have a strictly smaller $\text{dev}_{\text{best39}}$, including $\{\mu, b, t\}$ with dev = 0.005912 (§7.4). In the asymmetry ranking (by $\text{dev}_6 - \text{dev}_3$), it ranks 16th out of 84.

What was pre-specified. The pre-specified hypothesis was not "base 15 is special", but "some base $b \in [2, 40]$ yields an anomalously small dev". Base 15 turned out to be the minimum in this scan. The metric $p_{abs} = 10/84$ accounts for this search by construction: each triplet is allowed to choose its best base (best-of-39), which is why p_{abs} is 0.1190 rather than 1/84. Pre-registration is documented in `nb10_pvalue.md` §1–§2 (timestamp 2026-11-11 03:40) before the run.

7.4 Look-Elsewhere Effect

The most "grid-like" triplet under best-of-39 is $\{\mu, b, t\}$ at base 40 with a deviation of 0.005912, which is **4.76 times** better than the lepton signal (0.028148). This is a direct consequence of the look-elsewhere effect: scanning over 39 bases allows random triplets to "try" all bases.

7.5 Quarks

For quarks: MC mean at base 40 = 0.178335 ± 0.0146; deterministic minimum $\text{dev}_{\text{best39}} = 0.174249$. No base yields a deviation below 5%. The best base 40 is unstable (the top-5 lie within ±std of the Monte Carlo over PDG errors: spread 0.0035 < std 0.0146). The negative control is clean.

7.6 Fixed Base 15 (Secondary Line, with Caveat)

If base 15 is fixed without multiplicity correction, $p_{\text{fixed}} = 1/84 = 0.0119$. **Caveat:** base 15 was found by scanning these same data, so p_{fixed} is invalidated by post-selection. It is not used in the abstract or the bottom line.

7.7 Relation to Koide

The Koide relation $K = 2/3$ [1] has been known since 1981 and is not a result of this project. The nearest "nice" fractions (1/32, 1/91) are post-hoc: for any measured value, close fractions can always be found. The real fact of 10^{-5} precision is Koide for measured masses ($K = 0.66666051$, deviation 6.2×10^{-6}).

The 2.8% and Koide are not independent: integer exponents (0, 2, 3) yield $K = 0.65591603$; fractional parts (1/32, 1/91) yield $K = 0.66666378$ — exactly the degree of freedom needed to reach $K = 2/3$. The agreement "2.8% + Koide" is one piece of evidence in two formulations, not two independent ones.

7.8 Monte Carlo over PDG Errors

The Monte Carlo over PDG uncertainties (seed 42, $N = 500$) is a **stability test against measurement errors**, not a test of statistical significance. Result: quarks 0.178335 ± 0.0146, leptons $0.028148 \pm 1.61 \times 10^{-5}$. Significance is determined exclusively by the permutation test.

On the std at bases 16 and 4: the range 10^{-9}–10^{-8} is confirmed by the actual run (5.24×10^{-9} and 1.05×10^{-8}). The reason is structural: at these bases, the contribution of the mass perturbation to the three pairwise terms cancels pairwise to first order, unlike base 15, where they align.

Status of base 15: exploratory, not established. The framework does not depend on this hypothesis. Negative controls are preserved. The primary criterion (best-of-39) is not established ($p_{abs} = 10/84 = 0.1190$); the secondary criterion (fixed base 15, $1/84 = 0.0119$) is only valid with the post-selection caveat.

8 Falsification Criterion

The thesis of the separation of structure and data would be falsified if:

1. A statistically significant base is found for quarks in $b \in [2, 40]$ using the same method as for leptons ($p < 0.01$);
2. Adding local unitary interactions (a) changes the classification of any sector from "structure" to "data", or (b) requires fitting a parameter previously reproduced without fitting;
3. The CP phase is derived from QCA rules without free parameters, which would move it from "data" to "structure".

None of the three has occurred. The base-15 observation for leptons does not reach the general significance threshold of $p < 0.01$ (its $p_{abs} = 0.1190$ is far from it). This is consistent with the approach: base 15 was an exploratory observation, not a structural derivation.

9 Limitations

1. QCA is an approach, not a "theory of everything".
2. All successful results reproduce known physics. No new predictions are made; $\Sigma m_\nu = 0.0608$ eV (nb13) is a consistency check with the DESI DR2 upper bound [6], not a prediction.
3. Yukawa couplings are not explained.
4. The interpretation of the "4D operator" is philosophical.
5. **Base 15:** $p_{abs} = 10/84 = 0.1190$**, status not established**; the hypothesis is exploratory, and the framework does not depend on it. The look-elsewhere ratio is 4.76. The base-15 analysis serves as a methodological demonstration of falsifying numerical hypotheses, not as a physical discovery.
6. Neutrinos and constants do not follow the base-15 grid.
7. Three generations: the mechanism is exact in 1D, 2D requires overlap fermions; the number $N = 3$ is input.
8. CP violation: the framework can accommodate it, but the phase δ is input.
9. 2.8% and Koide are not independent (fractional exponents are a consequence of $K = 2/3$).
10. **The look-elsewhere factor 4.76 is a lower bound.** It accounts only for the 39 bases in nb10, not for hypotheses tested in v1–v13 (Koide, SVD, DESI, alternative normalizations). The true multiplicity correction is strictly larger than 4.76, so $p_{abs} = 0.1190$ **is a lower bound, not a full correction**. This makes the base-15 conclusion even less established, not more.
11. Robustness checks regarding the MC sample size (N=500 vs N=1000) and the random seed (42, 43, 44) yield identical rankings and deviations within the stated errors; the mean-vs-max aggregation is left for v15.

12. Multi-center cosmology is not tested. The approach is neutral to initial conditions (single-center vs. multi-center initialization).

13. Bounce cosmology and pre-inflationary phases are not tested.

14. Gravity is not tested. The emergence of $1/r^2$ from the QCA lattice is a task for a separate notebook (nb20, planned).

15. SM chiral left-handed doublets are not implemented. Overlap fermions (nb18 Part C) provide the technology, but the specific embedding of $SU(2)_L$ doublets is not done.

10 Conclusion

The "structure and data" approach is confirmed at 8 distinct levels of structure (nb01–06, 08, 18), 4 consistency checks with quantum mechanics (nb12, 15–17), 5 negative controls (nb07, 11, 14, 18A, 19 T0–T2), and 4 data classifications (nb09, 10, 13, 19 T4–T5).

Main result of v14.4. The separation of structure and data does not depend on the base-15 hypothesis.

- **8 distinct structural sectors** (Dirac, U(1), SU(2), mass-independence, Higgs, honeycomb lattice, 3D wall, 3 generations) — derived from QCA.
- **4 checks of Class B** (CPT, $c = 1$, entanglement, Pauli) — consistent with quantum mechanics.
- **5 negative controls of Class C** — the classification is non-trivial.
- **4 data classifications** (quarks, base scan, neutrinos, CP phase) — parameters are not derived.
- **Base 15** — an exploratory observation, **not established** ($p_{abs} = 10/84 = 0.1190$, lower bound; look-elsewhere ratio 4.76). The framework does not depend on it.

The thesis "structure is derivable, data is input" is **falsifiable**: the three criteria in Section 8 define the conditions for refutation. None of them has been realized. The approach is neither a "theory of everything" nor the simulation hypothesis; it is a classification of what follows from local unitarity and what must be given from outside.

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